Tip of the Iceberg: How Much Do Tips Bunch at Reporting Thresholds?

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Abstract

We study the importance of bunching in the context of tip-income reporting by workers at full-service, single-unit restaurants in the United States. Using tax reports at both the individual and the employer levels, we show that reported tip income varies with minimum-wage laws that provide an incentive for tipped workers to report some, but not necessarily all, of their tips. As a result, reported tips bunch at the minimum required threshold. We quantify missing tips due to bunching at nearly $63 million per year in 2018 dollars, on average over the period 2005-2018. Bunching is stronger for jobs at small employers and in the earlier part of the time series and declined monotonically from 2010 to 2018. Using restaurant-level revenue data, we also estimate the total value of unreported tips assuming an average tip rate of 12%. We find that tips are missing throughout the distribution. All told, missing tips exceed $4 billion per year, implying that bunching explains only 1.5% of all missing tips.

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1 Introduction

Bunching has been observed in many settings, often involving income that is easy to conceal, such as self-employment income or income paid in cash. We examine bunching in a novel setting: U.S. restaurant workers' tip income. The incentive behind bunching of tip income derives from the structure of minimum-wage laws in most states, in which an employer can pay a sub-minimum wage to tipped workers as long as they earn enough tips to make up the difference between the subminimum wage and the standard minimum wage; this difference is called the *tip credit*. We document that bunching does indeed occur at the tip-credit threshold: approximately 30% of jobs reporting tips in the neighborhood of the reporting threshold likely earn more in tips than they report. Using a two-part nested randomization approach, we estimate missing tips due to bunching at $62.8 million per year in 2018 dollars, on average over the period 2005–2018. We show that bunching is more common for jobs at small than large employers, possibly because small restaurants are managed less formally and are subject to fewer regulations.

Although bunching is both economically and statistically significant, it is not the primary cause of tip under-reporting. In the second part of the paper, we calculate total missing tips using restaurant revenue data. Assuming an average tip rate of 12%, we estimate missing tips for any restaurant whose workers, collectively, report tips amounting to less than 12% of revenue. By this calculation, missing tips exceed $4 billion per year, implying that missing tips due to bunching account for only 1.5% of all missing tips. Tips are under-reported throughout the distribution, not just near the minimum-wage reporting threshold.

The analysis combines several large administrative datasets on restaurants and workers, covering U.S. full-service single-unit restaurants from 2005 through 2018. We use the Longitudinal Business Database (LBD) to identify restaurants and obtain their annual revenues. We link workers to restaurants using Form W-2 filings and obtain wage earnings and tips at the job level. Our data allow us to track virtually all jobs at full-service single-unit restaurants over this period, a panel that includes 9.5 million workers at over 400,000 unique

restaurants.

Our bunching analysis is related to several recent papers showing bunching of reported income at minimum-wage thresholds. Bíró, Prinz, and Sándor (2022) document that reported incomes of workers in Hungary often bunch at the minimum wage, when, in fact, a substantial share earn at least twice that amount. Gavoille and Zasova (2023) also document substantial under-reporting of wage income among workers in Latvia. Most closely related to our paper, Feinmann, Hsu Rocha, and Lauletta (2024) use data from Brazil to document that a quarter of formal-sector workers report getting some of their wages "under the table," and demonstrate that both employers and employees are sensitive to reporting incentives. We focus on the United States and examine an infrequently studied source of income: tips paid to restaurant workers.

In the U.S., evidence of bunching of reported income has previously been found only in self-employment income and not in wage and salary income (see. e.g., Saez, 2010; Imboden, Voorheis, and Weber, 2023). A theme that has emerged in the literature on bunching of self-employment income is the need to distinguish between behavioral responses to incentives—e.g., taxpayers targeting a specific level of income that maximizes funds—from reporting responses—taxpayers reporting income that may not reflect actual income (le Maire and Schjerning, 2013; Mortenson and Whitte, 2020). In our setting, behavioral responses are generally implausible. This is because the bunching we document is of a ratio (tips as a share of earnings) and is therefore not sensitive to hours worked, which is the most obvious behavioral variable that workers are able to manipulate. In addition, tips are not known by the server in advance of providing the service, so they are difficult for servers to manipulate.

The rest of this paper is organized as follows. Section 2 describes the institutional and legal environment that motivates our empirical exercises. Section 3 describes our data sources, with some omitted details presented in Appendix A. We show graphical evidence of bunching in Section 4. We discuss our bunching estimator and results, as well as our estimate of missing (unreported) tips due to bunching, in Section 5. Section 6 provides an

alternative estimate of missing tips using restaurant revenue and compares the two measures of missing tips. Section 7 concludes with implications for measurement and policy.

2 Institutional Setting

The primary incentive for workers to report tip income is a provision in the Fair Labor Standards Act (FLSA) that allows employers to pay tipped workers a lower (subminimum) hourly base wage, provided the workers earn enough tips to make up the difference between the subminimum wage and the standard minimum wage. Employers therefore have an incentive to ensure that their workers report that difference—the tip credit—in tips for every hour of work. For workers whose tips fall short of the tip credit, the employer's incentive is to ensure all tips are reported, because they reduce the worker's applicable minimum wage one for one. In many cases, the tip credit is substantial: In 19 states and the District of Columbia, the tip credit in 2018 exceeded 70% of the minimum wage. In most of these states, the hourly subminimum wage for tipped workers is \$2.13/hour and the standard minimum wage is \$7.25; if a worker earning this subminimum reports less than \$5.12/hour in tips, the employer is required to make up the difference.[1]

In states with tip credits, the state-level tip credit over our sample period ranges from 16.7% of the minimum wage (in New York in 2016) to 76.3% of the minimum wage (in Nebraska in 2016–2018). In states in which the federal minimum wages are binding, the tip

[1]This rule does not apply to all workers or at all restaurants. Seven states had no tip credit throughout the period we study: Alaska, California, Minnesota, Montana, Nevada, Oregon, and Washington. We also treat Hawaii as a non-tip-credit state. In Hawaii, workers need to earn several dollars in tips to be eligible for a very low (25 cent or 50 cent) tip credit. The FLSA, which specifies the federal minimum and subminimum wages and some worker rights, exempts enterprises that have annual gross volume of sales of \$500,000 or less and those with only one employee. However, workers handling credit-card transactions are deemed to be engaged in interstate commerce, which is covered under FLSA (U.S. Department of Labor, Wage and Hour Division, 2009, 2016). Consequently, the tip-reporting incentives for small restaurants are less clear than for larger ones, with FLSA coverage possibly varying by employee. In addition, workers are eligible for the lower minimum wage only if they earn at least \$30 a month in tips.

credit ranges from 58.6% (until July 23, 2007) to 70.6% (starting July 24, 2009).[2]

Restaurants also face a tip-reporting threshold. A provision in the Tax Equity and Fiscal Responsibility Act of 1982 requires restaurants report tips equal to at least 8% of revenue or face penalties (Newman, 1988). This creates an incentive for restaurants to ensure their workers collectively report enough tips to meet this 8% threshold. The 8% revenue-share rule only applies to restaurants with at least ten employees, collectively working a total of at least 80 hours, on a typical business day, and where tipping is customary (Internal Revenue Service, undated).

In addition to these incentives, starting in 2011, a provision of the Housing and Economic Recovery Act has required aggregate third-party network transactions, including tips, be reported to the Internal Revenue Service (IRS) on Form 1099-K.[3] Form 1099-K is filed by the credit-card processor and includes the gross amount processed through the network. Research has shown that this type of third-party reporting can increase income reporting. For example, Garin, Jackson, Koustas, and Miller (2023) show that self-employed workers who receive payments from online platforms such as ride-sharing apps are more likely to report self-employment income on their tax returns if they receive a Form 1099-K from the platform. Although Form 1099-K does not separately report tips, it still may affect incentives to report tips accurately to the IRS for restaurants that accept credit and debit cards, due to increased transparency.

Apart from these reporting incentives, workers generally have an incentive to under-report their tips. Workers owe payroll taxes (7.65%) on all wages and tips, as well as, potentially, state and federal income taxes, all of which create an incentive to minimize tip

[2]The federal tip credit was 63.6% of the minimum wage from July 24, 2007 to July 23, 2008, and 67.5% from July 24, 2008 to July 23, 2009.

[3]Similar to the other reporting incentives, Form 1099-K is only required if the network processed a minimum number of transactions and a minimum value. From 2011 to 2018, the thresholds for a 1099-K were $20,000 in reportable transactions and more than 200 such transactions (Internal Revenue Service, 2013, 2019b). Some states have lower thresholds; for example, in Vermont starting in tax year 2017, the value threshold was reduced to $600 and the transaction count threshold was removed altogether (Vermont Agency of Administration, Department of Taxes, 2017).

reporting. The reporting disincentives operate at all levels of tipped income.

3 Data Sources

We combine several large administrative datasets on restaurants and their workers to create a comprehensive panel of U.S. full-service single-unit restaurants from 2005 through 2018.[4]

We start with data on restaurants from the Census Bureau's Business Register (BR), Longitudinal Business Database (LBD), and Economic Census. Each of these datasets includes detailed North American Industry Classification System (NAICS) codes that allow us to distinguish full-service restaurants from other restaurants. The Economic Census, conducted in 2007, 2012, and 2017, includes questions directed at restaurants about service levels, which allow us to refine the assignment of industry designations.

Once we identify records associated with full-service single-unit restaurants in the BR and LBD, we use data from IRS Form W-2 records to identify workers at those restaurants. Filed annually by employers, Form W-2 provides a breakdown, by job-year, of wage earnings and reported tips. We distinguish between a worker and a job: the former is an individual; the latter is a worker-employer pair. Our analysis in this paper is at the job-year level. Workers can have multiple restaurant jobs in a year.

The datasets also include other pertinent information about restaurants. First, the business-level datasets include the business address, which we use to determine the applicable minimum wages. We link minimum wages based on state, and in some cases city or county, as well as business size. (More details on our minimum-wage data are in Appendix A.) Second,

[4]We restrict our sample to single-unit restaurants because our data linking workers to firms do not assign a job to a particular restaurant location. Consequently, for jobs at multi-unit (chain) firms, it is not always possible to know which minimum-wage laws apply, nor is it possible to know which restaurant locations within the firm report tips. We also have annual revenue at the restaurant level only for single-unit firms. The single-unit designation is based on ownership information; therefore, a single-unit restaurant may be a franchise of a familiar name brand if the restaurant's owner does not operate other restaurants. In 2017, single-unit restaurants accounted for about 84% of full-service restaurants, 62% of full-service restaurant employment, 63% of full-service restaurant payroll, and 65% of full-service restaurant revenue. These statistics can be derived from data at U.S. Census Bureau (2017).

for most single-unit restaurants, the LBD includes annual revenue, also derived from IRS tax filings. We use restaurant revenue in Section 6 to calculate the total value of tips across all jobs for a given employer, assuming a 12% tip rate on the average dollar of revenue.

For more details about the data, see Appendix A.

4 Descriptive Evidence of Bunching

If a worker in a job with a two-tier minimum wage reports tips exactly equal to the applicable tip credit in every hour of every shift throughout the year, her ratio of tips to total earnings equals the ratio of the tip credit to the minimum wage:

$$\frac{\text{tips}}{\text{wages} + \text{tips}} = \frac{\text{tip credit}}{\text{minimum wage}}$$

To examine reporting incentives, we normalize the tip share of earnings by the tip-credit ratio:

$$\frac{\left(\frac{\text{tips}}{\text{wages+tips}}\right)}{\left(\frac{\text{tip credit}}{\text{minimum wage}}\right)}$$

This normalization allows us to pool jobs in states with varying levels of tip credit, and across all years, despite variation in both the federal and state-level tip credits.

Figure 1 shows the distribution of the normalized tips-to-earnings ratio for jobs with reported tips in all states with a tip credit.[5] The bunching of the normalized tips-to-earnings ratio at 1 is consistent with many workers employed in these jobs misreporting their tips to their employer and the IRS.

One reason for bunching could be tip pooling at the restaurant, whereby tipped workers put some or all of their tips into a common pool from which they are redistributed. Such

[5]We show the distribution using a kernel density because Census Bureau disclosure-avoidance rules prohibit us from displaying a histogram. Because the kernel density graph smooths over nearby points, bunching may appear less concentrated than it actually is.

a practice could move both highly tipped workers and poorly tipped workers closer to the tip credit.[6] Unfortunately, we are not aware of any data on tip pooling at the restaurant level that would enable us to assess how much of the bunching we observe is due to pooling. However, our exercises in the next section suggest that pooling cannot be the only, and likely not even the primary, reason for the observed pattern.

Larger restaurants may be more formally managed or have more credit-card transactions, and workers in larger restaurants may, therefore, be less likely to bunch at the tip credit. In addition, the Tax Equity and Fiscal Responsibility Act of 1982 provision that requires restaurants to report tips equal to at least 8% of revenue applies only to restaurants with more than ten workers on a typical business day. This additional tip-reporting incentive could attenuate the impact of the tip-credit reporting incentive, reducing observed bunching at the tip credit. Small restaurants, with fewer than ten workers per day, are exempted from this requirement. Although we do not observe the number of employees on a typical day, we do know how many workers have passed through the restaurant over the course of the year from Form W-2 filings. We use a cutoff of 20 Form W-2 employees to designate small vs. large employers. Figure 2 shows the distribution of the normalized ratio of tips to earnings for jobs at small employers, with fewer than 20 employees, compared to large employers, with 20 or more employees. Bunching is clearly more pronounced for jobs at small employers.

Bunching decreased following the implementation of the Housing and Economic Recovery Act provision on third-party reporting of credit-card totals to the IRS in 2011. Figure 3 shows the density of the normalized tips-to-earnings ratio separately for the first part of the sample, 2005–2010, and the second part, 2011–2018. The distributions to the left of the

[6]Eisenberg and Williams (2015, p. 394) claim that most restaurants implement tip pooling. Mandatory tip pooling is allowed in most states and voluntary pooling is allowed in all states. During most of the study period, mandatory tip pools could include only workers who interacted directly with customers and could not include managers. This rule was relaxed in March 2018, near the end of our sample period (Federal Register, 2021, p. 52975). Mandatory tip pooling may not eat into the tip credit and may not exceed a "customary and reasonable" percentage of an employee's tips, which the U.S. Department of Labor defined as no more than 15% until 2011 (Eisenberg and Williams, 2015, p. 398; that threshold was relaxed following litigation). Informal sharing of tips, which is unlimited and unregulated, may include kitchen staff and managers.

bunching point look very similar in the earlier and later periods. However, the distribution to the right of the bunching point is lower for the earlier period and excess bunching appears to be larger. In the next section, we formalize the calculation of excess bunching and more rigorously measure changes over time.

5 Estimation of Bunching

5.1 Methodology

We adapt the methodology proposed by Chetty, Friedman, Olsen, and Pistaferri (2011, hereafter CFOP) to quantify bunching in our data.

We begin by grouping observations into bins of size 0.01 based on their value of the normalized tips-to-earnings ratio. We calculate each bin's share of the distribution and estimate a bin-level regression of the observation share on a sixth-order polynomial of the bin number, using bins in the range $[0.6, 0.95) \cup (1.05, 1.4]$, which is to the left and the right of our bunching window of $[0.96, 1.05]$. We use the predicted (fitted) value from this regression to assign a counterfactual share to each bin in the bunching window; this counterfactual represents the share of observations that bin would have had if there had been no incentive for servers to report tips in a particular proportion to their wages. Replacing the actual shares with predicted shares leaves some of the mass of the original distribution unaccounted for; to correct for this reduction, we spread the missing mass on the region $[1.06, \infty)$ in proportion to the shares already in each bin, implicitly assuming that all workers originally in the bunching window under-reported their tips. Because the new distribution exhibits a discontinuity at 1.06, we iterate this procedure to convergence. This creates a final counterfactual distribution with no bunching.[7]

[7]We follow CFOP closely but differ in two respects. First, we use a sixth-order polynomial instead of a seventh-order polynomial because our bunching window is slightly smaller than theirs. Second, whereas CFOP distribute observations equally among all the bins to the right of the bunching window, we distribute in proportion to the observed size of each bin.

To create a point estimate of bunching, we calculate the difference between the share of observations in the bunching window using the actual distribution and the share using the counterfactual distribution. To generate a standard error, we use the error terms from the bin-level regression. We assign these error terms with replacement to the bins in the estimation window and recalculate. We repeat this procedure 100 times and use the standard deviation of the distribution of the resulting bunching estimates as the standard error of our estimate.[8]

We report excess bunching as a share of the reported observations in the bunching window. This differs from CFOP in two ways. First, we calculate the excess bunching for the bunching window as a whole, instead of bin-by-bin. Second, our denominator is the number of reported observations in the bunching window, rather than the counterfactual observations in an average bin in the bunching window. Using a larger denominator mechanically reduces our estimate of bunching.

5.2 Results

Figure 4 shows the original ratio of ratios and the revised version with no bunching. The mass in the bunching window $[0.96, 1.05]$ has been shifted to the right, creating a smooth density function. Excess bunching, defined as the share of observations in the bunching window that we shift to the right, is 19.82%. Because the kernel density graph smooths over nearby points, the true bunching window is likely smaller than what we see graphically, implying that this calculation under-estimates excess bunching.

[8]Starting from the point estimates of the final no-bunching counterfactual distribution, we add one of the predicted residuals from the bin-level regression to each of the bins in the estimation window (excluding the bunching window), where that predicted residual is selected at random with replacement from the full set of residuals. After applying this perturbation, we re-estimate the bin-level regression (repeating it to convergence, as above) and calculate the difference between the share of observations in the bunching window using the actual distribution and the share using the predicted distribution. Each iteration of this exercise generates a point estimate of bunching. We use the standard deviation across 100 iterations.

The CFOP measure of bunching provides an answer to an alternative thought experiment. Suppose all the excess observations in the bunching window were concentrated at the reporting threshold, i.e., in our setting, at $\frac{\left(\frac{\text{tips}}{\text{wages+tips}}\right)}{\left(\frac{\text{tip credit}}{\text{minimum wage}}\right)} = 1$. As a share of the counterfactual (no-bunching) distribution at that point, how much would that excess be? If we replicate their calculation in our data, bunching in Figure 4 is approximately 250%.

The bunching of the normalized tips-to-earnings ratio cannot be explained by any behavioral response. Unlike self-employment income, which can reach a kink point in the tax code by manipulating actual incomes as well as changing their reporting behavior (Saez, 2010; le Maire and Schjerning, 2013), it is virtually impossible for tipped workers to reach the bunching point by manipulating actual tips. This is because, first, the ratio of tips to earnings is not a function of hours of work, the most obvious variable over which workers have some control. Second, because tips are determined *ex post* by the customer, they are much harder for a worker to manipulate through behavior, such as providing more or less courteous service (Lynn, 2001).

We noted earlier (Figure 2) that bunching appears to vary by employer size. We report bunching for four employer-size bins in Table 1. There is no meaningful difference between bunching at employers with fewer than 10 employees and at employers with 10–19 employees; in both cases, we estimate that around 30% of observations in the bunching window belong elsewhere in the distribution. Above 20 employees, however, we do see some variation. Bunching at employers with 20–39 employees is slightly lower, at about 26%. For jobs at employers with 40 or more employees, bunching is just 15%, consistent with a greater formality, and possibly a greater reliance on charge cards, at larger restaurants.

Bunching also varies over time, moving with minimum-wage rules. Consistent with Figure 3, Figure 5 shows our estimate of excess bunching by year. Bunching accounts for approximately 20% of observations in the bunching window in 2005 and 2006, after which it increases gradually, reaching 25% by 2010. This increasing trend is likely due to increases in the federal minimum wage. The federal minimum wage increased three times over this

period, in 2007, 2008, and 2009. These increases occurred midyear (effective July 1); the tipped subminimum wage did not increase, so the tip credit increased each time. As a result, the tips needed to meet the tip credit increased, which had the effect of increasing bunching at the tip credit.[9] Although not all jobs in the sample were subject to the federal minimum wage, many were: in 2008, 30% of jobs at full-service single-unit restaurants subject to any tip credit were subject to the federal tip credit.

After 2010, bunching decreased gradually, falling to 11.5% by 2018. This decrease may reflect two trends. First, as noted earlier, the combination of increased card payments and the 2011 implementation of the Housing and Economic Recovery Act provision requiring third-party reporting of credit-card totals have increased transparency and made it easier for workers and their employers to track tips. In addition, the erosion of the tip credit in real terms may have decreased its salience over time. There is no reason, in contrast, to believe that tip pooling decreased dramatically over this period, casting doubt on pooling as an explanation for bunching at the reporting threshold.

It is perhaps surprising to see no discontinuity in bunching in 2011, the year in which third-party reporting of credit-card totals started. One reason for this could be that businesses were not aware of the change or its implications; Zwick (2021) notes that small firms may be less sophisticated in understanding the tax code, and this lack of sophistication may extend to other regulations. Another possibility is that tip reporting improved throughout the distribution but not necessarily around the bunching point.

To examine how changes in the tip credit affect the distribution, we use two case studies, one from New York, and one from Texas.

The tip credit in New York increased from 37.5% in 2014 to 42.9% in 2015, and then dropped to 16.7% in 2016.[10] Figure 6 shows the distribution of the (non-normalized) ratio of

[9] We walk through how increases in the minimum wage can affect bunching in Appendix B.

[10] The 2014 minimum wage in New York was \$8, with a \$5 subminimum for tipped workers, so the tip credit was $(^{8-5}/_{8} = 0.375)$. In 2015, the minimum wage increased to \$8.75 and the tipped subminimum wage remained at \$5. In 2016, the minimum wage increased to \$9 and the tipped subminimum increased to

tips to earnings for jobs in New York in 2014, 2015, and 2016. It is clear that the bunching point moves with the tip credit: when the tip credit increased in 2015, the bunching point increased with it; when it decreased in 2016, the bunching point decreased as well. As a share of the observations in the bunching window, excess bunching in New York decreased from 27.6% in 2014 to 22.7% in 2015, and then increased rather dramatically to 42.4% in 2016. These changes are largely due to the change in the denominator: there are relatively few observations in the 2016 bunching window $[0.16, 0.175]$, so the bunching is larger as a share of that window.

Perhaps more strikingly, the entire distribution of tips as a share of earnings shifts with the tip credit. For example, when the tip credit decreased in 2016, the probability of a worker reporting tips equal to more than 50% of her earnings—a level above any recent tip credit in the state—decreased dramatically. This finding highlights that the salience of the tip credit extends well beyond the bunching threshold and suggests that under-reporting of tips may not be limited to observations near the reporting threshold.

Unlike New York, Texas does not set its own minimum wage; therefore, the federal minimum wage applies in Texas. The federal tip credit increased midyear in 2007, 2008, and 2009, which muddies our annual analyses for these years; instead, in Figure 7, we plot the distribution of the (non-normalized) ratio of tips to earnings for tipped jobs in Texas using data from 2005, when the tip credit was 58.6%; and 2010 and 2018, when it was 70.6% by 2010.[11] Here the picture looks rather different from New York: bunching decreased from 18.5% in 2005 to 14.8% in 2010.[12] Our point estimate of bunching is slightly negative (-1.4%) in 2018, but it is not statistically different from zero. As with New York, we observe

$6.50. In 2017, New York started phasing in minimum wages that vary by geography and employer size, so there is no longer a single tip credit for all full-service restaurant jobs in the state.

[11]The federal subminimum wage was $2.13 throughout this period. The minimum wage was $5.15 until July 2007; it increased in July 2007, July 2008, and July 2009. Since July 2009 the federal minimum wage has remained at $7.25.

[12]It is difficult to spot the bunching in 2010 with the naked eye, because it is so close to the natural mode of the distribution.

the entire distribution shifting with the tip credit.

These results imply a tight relationship between reported tips and the tip credit, but that relationship should not be interpreted to mean that actual tips increase with the tip credit. Rather, we attribute changes in reported tips following minimum-wage changes mostly to increased tip reporting due to the need to cover the expanded tip credit. In contrast, a recent paper by Talesara and Wickard (2025), which uses IRS tax filings and finds, like us, that reported tips increase with the minimum wage, argues that this relationship reflects an increase in patrons' tip rates. Talesara and Wickard (2025) show evidence that neither changes in revenue per worker nor changes in tip pooling are sufficient to explain this increase in tips. Instead, they conclude that restaurants are able to incentivize an increase in patrons' average tip rate. In contrast, our finding of strong bunching at the tip credit leads us to conclude that reporting changes on the part of workers, incentivized by hourly wage requirements, cause the differences in the observed tip rate.

5.3 Missing Tips Due to Bunching

Excess bunching implies that tips are under-reported. However, translating excess bunching into a dollar measure of missing tips due to bunching requires making consequential assumptions. First and most important, we argue that all of the observed response to the reporting threshold is on the reporting margin; in other words, there is no behavioral response that reduces tipped workers' actual tip income. Our task would be much more difficult if we believed adjustments occurred on both the reporting margin and on the earnings margin (see, e.g., Aghion, Gravoueille, Lequien, and Stantcheva, 2017). Second, bunching tells us how many jobs are over-concentrated in the bunching window but not which jobs they are, nor where in the distribution they should be. To determine the dollar value of missing tips, we need to decide which jobs are under-reporting and how much each of them should have reported in tips. To do this, we use a nested randomization approach.

We use our point estimate of excess mass in each bin in the bunching window to determine the number of jobs that should have more tips assigned. We then use two nested randomizations to choose, first, which jobs to move, and second, where to move them. For the first randomization, we select which jobs in each bin will get more tips in proportion to the excess mass in that bin. We repeat the first randomization ten times. For each of these randomizations, we redistribute the selected jobs the same way we created the counterfactual distribution in Section 5.2: randomly assigning them to bins in the range of $[1.06, \infty)$ in proportion to the shares already in each bin. Once each job selected to move has been assigned a new bin, we calculate the level of tips necessary for the job to belong to the new bin, using the job's actual minimum wage and tip credit; we infer missing tips by subtracting reported tips from the counterfactual tips. We repeat the random assignment of new locations ten times for each of the random draws of jobs in the first randomization, for a total of 100 iterations. We use the mean and standard deviation of these 100 iterations to estimate missing tips due to bunching.

This calculation results in a non-negative value of bunching-induced missing tips for each job in our data. Zero missing tips are assigned to all jobs outside the bunching window and to jobs in states with no tip credit. Jobs in the randomly selected subset from the bunching window are all assigned positive missing tips because they are all shifted to the right of the bunching window. The remaining jobs in the bunching window, which are not reassigned, have zero missing tips.

The assumption that jobs in the bunching window should be in the range of $[1.06, \infty)$ in proportion to the shares already in each bin clearly matters for the results. First, this assumption implies that workers never over-report their tips under pressure from employers. To the extent that over-reporting does happen, we are over-estimating missing tips. Second, we do not bound the window into which jobs are reallocated from above; had we done so, our estimate of missing tips would be lower. However, because we reassign jobs in proportion to the reported distribution and the underlying distribution is bell-shaped, there are relatively

few reassignments to the far right of the distribution; most of the mass is to the left of 1.5. Saez (2010), in contrast, reassigns observations in equal proportion throughout the distribution. Had we done that, our estimate of missing tips would be higher.

For the full sample, our estimate of missing tips due to bunching is about $63 million per year. The standard deviation across the 100 iterations of the calculation is only $155,000, implying a coefficient of variation of about 0.2%. One reason for this relative lack of variation is that the number of observations from the bunching window that are selected to be reassigned is large enough (over 550,000) that laws of large numbers apply. The second randomization also has relatively little variation because we use the distribution of reported tips to determine the probability distribution for reassignment, resulting in the majority of observations being redistributed in a relatively narrow range above the bunching point.

We also calculate missing tips due to bunching separately for each of the four employer-size categories discussed above. Nearly half of missing tips due to bunching, or about $29 million per year, are at the largest employer size: employers with 40 or more Form W-2 workers. This is because, although bunching is relatively modest in jobs at large employers, these large employers account for two thirds of all jobs in our sample.

6 Tip Under-Reporting Beyond the Bunching Window

6.1 Motivation

In this section, we take a step back from bunching and use data available at the restaurant level to estimate missing tips over the entire distribution. This approach allows us to include jobs in states without a tip credit, which are excluded from the previous analysis, as well as jobs at restaurants with no reported tips. Our goal is to examine the extent to which tip under-reporting is concentrated around the bunching window.

We have a strong prior belief that tipping occurs at virtually all full-service restaurants

in the U.S.[13] We also have prior beliefs on the aggregate value of tips: the average tip rate is likely in the range of 10–20% of the bill. Azar (2020) cites various sources that recommend a tipping range of 15–25% of the check amount. Public data we obtain from two consumer surveys and a payment application are on the lower end of this distribution, much closer to 15% than 25%. The first consumer survey, FoodAPS, conducted by the U.S. Department of Agriculture, references nearly 2,000 meals, almost all of which took place in 2012; the average tip rate in that survey is 14.6% (public-use data files and documentation are available at U.S. Department of Agriculture, Economic Research Service, 2016). In another consumer survey from late 2019, shortly after the end of our sample period (but, importantly, before the beginning of the Covid-19 pandemic), 82% of respondents said they regularly tip waitstaff at restaurants (TD Ameritrade, 2020).[14] Among those who did leave a tip, the median reported tip rate is in the 19–25% bin, with only 11% tipping at higher rates. Finally, payment data from the Square payment app, published in 2014, when the app was used mostly by small businesses, show that about 57% of restaurant transactions include tips; among transactions that were tipped in the application, the average tip rate was approximately 16.5%, but some of the transactions without a tip were likely tipped in cash (Square, 2014).

Despite their very different minimum-wage laws, there is no evidence that patrons behave differently in states with and without a tip credit. Square (2014) shows that the average tip rate does not vary much by state tip credit: it was 16.4% for transactions with tips in the seven states that had no tip credit, and 16.3% on average in the 17 states that used the federal minimum wage in 2014. In addition, although travel and etiquette guides highlight differences in tipping norms between the U.S. and European countries, they do not single out states without a tip credit to suggest that patrons tip less there (Azar, 2020).

[13]There are some full-service restaurants that do not allow tipping, but casual observation suggests that they are exceedingly rare.

[14]The survey included approximately 1,000 U.S. adults ages 23 and older who had assets of at least $10,000.

6.2 Methodology

To calculate missing tips, we choose a conservative measure of 12%, which can be rationalized with a 15% tip rate and a 20% rate of "stiffing" (leaving no tip). For each restaurant r and year t, we calculate

$$\text{Missing}_{rt} = \max\{0, 0.12 \cdot \text{Revenue}_{rt} - \text{Tips}_{rt}\}$$

where Tips_{rt} is the aggregate value of tips reported across all jobs at the restaurant. If the restaurant's workers collectively report tips accounting for more than 12% of reported revenue, we classify it as having no missing tips; otherwise, we use the difference between 12% of revenue and actual tips reported.[15]

Because this calculation is at the restaurant level, it does not require assigning missing tips to specific jobs. However, assigning tips to specific jobs allows us to visually compare the resulting distribution of the normalized tips-to-earnings ratio to the reported and no-bunching distributions. We do the assignment as follows. Within each restaurant with tipped workers, we assign missing tips to jobs in proportion to the tips they do report, implicitly assuming that all jobs in a restaurant-year under-report their tips at the same rate. (In Basker, Foster, and Stinson, 2024, we show that the extensive margin for tip reporting is very accurate.) There are also some restaurants with a full-service NAICS code that do not report any tips. A hand audit we report in Basker, Foster, and Stinson (2024) suggests that about three quarters of these non-tip-reporting restaurant-year observations are actually full-service restaurants that should have reported tips. We randomly select 75% of restaurants with a full-service NAICS code that have no reported tips and assign them tips equal to 12%

[15]To the best of our knowledge, we are the first to estimate industry-level missing tips by aggregating restaurant-level missing tips. Azar (2003) and Basker, Foster, and Stinson (2024) use aggregate data and assume a tip rate of 15% to estimate missing tips. That approach significantly undercounts missing tips, because restaurants with higher reported tip rates cancel out some of the missing tips at restaurants with lower reported tip rates. Our measure may still undercount missing tips if the actual tip rate is higher than 12% or if restaurants under-report revenues.

of their reported revenues; we then allocate these tips equally among a randomly selected sample of half the workers at the restaurant.[16]

6.3 Results

Our calculations are shown in Table 2. The third row shows jobs at full-service single-unit restaurants in tip-credit states with at least some reported tips. Collectively, reported tips at these restaurants averaged about $9.85 billion per year. We estimate that an additional $2.68 billion are unreported annually at these restaurants, comprising about 21% of total tips. This calculation puts the $63 million missing tips due to bunching in context; bunching accounts for just 2.3% of all missing tips at restaurants with at least one tipped worker in states with tip credits. Even if we have significantly under-estimated missing tips due to bunching, it is clear that missing tips extend far beyond the minimum-wage bunching point.

The share of tips that go unreported is correlated with firm size. Larger firms in states with tip credits not only have less bunching of tips but also lower shares of missing tips overall. At small employers (with fewer than ten employees), missing tips account for 52% of all tips, compared to 39% of tips at employers with 10–19 employees, 27% of tips at employers with 20–39 employees, and just 13% of tips at employers with 40 or more employees. These reporting patterns are consistent with our hypothesis that servers at larger firms report more tips due to a combination of more formalized record keeping and higher usage of credit cards by customers. Large firms are the only group that average close to 12% tips per revenue (11.6%). The smallest firms, in contrast, report only 6% tips per revenue.

Assigning these aggregate missing tips to specific jobs changes the distribution of the normalized tips-to-earnings ratio, as shown in Figure 8. The revenue-based counterfactual distribution, shown in light gray, is shifted to the right, underscoring that there is under-

[16]Validation check on industry codes that we report in Appendix A are consistent with an accuracy of 75% in industry coding. Among full-service single-unit restaurants with reported tips, approximately half of workers report tips (Basker, Foster, and Stinson, 2024).

reporting not just near the bunching point but also in jobs with very low and very high levels of reported tips.[17] This figure also clarifies the importance of the assumptions we use to derive missing tips due to bunching. Because the excess bunching is small relative to the full distribution, even if we had reassigned tips from the bunching window more evenly throughout the distribution, we would not have come close to the revenue-based counterfactual.

In addition to restaurants that under-report tips, we estimate that another $1 billion in tips are missing annually due to non-reporting on the extensive margin: restaurants where no workers report earning any tips. As noted above, we assume that 25% of restaurants with no reported tips are actually not restaurants at all—and should have no reported tips—but that the majority are full-service restaurants that should have reported tips. Combining missing tips across all categories, we estimate a total of $3.7 billion, or 27% of all tips, go unreported annually in tip-credit states.

Using revenue to calculate expected tips also allows us to quantify reported and missing tips at full-service single-unit restaurants in non-tip-credit states using the same methodology. Workers at these restaurants collectively reported $2.33 billion annually in tips and failed to report an additional $1.42 billion, which translates to 38% of total tips not being recorded. Figure 9 shows the distribution of reported tips and counterfactual tips as a share of earnings in non-tip-credit states. We do not normalize the tips-to-earnings ratio in this sample because there is no tip credit by which to normalize. As with the tip-credit states, the revenue-based tip calculation shifts the entire distribution of the tips-to-earnings ratio to the right.

[17] Our assignment of missing tips to workers does not condition on where the workers fall in the distribution of the tips-to-earnings ratio. Some restaurants with missing tips have workers at different points in the distribution, and our assignment algorithm mechanically shifts all of them to the right, even if, in reality, only some of these workers under-report their tip earnings. Our examples of New York and Texas in Section 5.2, however, show that the entire distribution of the tips-to-earnings ratio is sensitive to tip-reporting incentives. Like us, Talesara and Wickard (2025) find that the tip credit affects reported tips throughout the earnings distibution.

When we combine the missing tips from all restaurants that report at least some tips in both non-tip-credit and tip-credit states ($1.42 billion plus $2.68 billion annually), missing tips due to bunching account for only 1.5% of all missing tips. We believe this estimate is conservative because tips are likely to represent more than 12% of revenue at many (but not all) restaurants. The fact that there are substantial unreported tips in states with no tip credit underscores that unreported tips are not a feature uniquely associated with the minimum-wage reporting threshold. At the same time, our calculation that more than a third of tips are unreported at restaurants in states with no tip credit, compared with only a fifth of tips at restaurants with a tip credit, implies that the tip credit does provide a powerful reporting incentive.

Combining missing tips across all categories, we estimate a total of $5.1 billion, or 29% of total tips on average are not reported each year. This figure is similar to Hurst, Li, and Pugsley's (2014) estimate that 25% of self-employment income goes unreported.

The finding that missing tips are not confined to the reporting threshold echoes recent findings in the self-employment literature. Garin, Jackson, and Koustas (2024) show that individuals who become eligible for Earned Income Tax Credit (EITC) rebates in the last week of a calendar year due to the birth of a child report higher annual self-employment earnings when they file tax returns for the year with the IRS than similar workers who do not become EITC-eligible, despite the fact that these earnings depend on past (and therefore sunk) labor-supply choices. Instead, the parent increases income reported to the IRS in a way that increases the refundable tax credit. Consistent with our findings, Garin, Jackson, and Koustas (2024) show that only a small share of this skewed reporting leads to bunching at kink points.

7 Concluding Remarks

Our goal in this paper is to better understand the nature of misreporting of tip income in the context of full-service, single-unit restaurant workers. We find that, on average over the period 2005 to 2018, unreported tips likely topped $4 billion annually in 2018 dollars. Approximately 1.5% of this amount is due to bunching at the minimum-wage reporting threshold; the remainder is due to under-reporting throughout the distribution. These estimates of missing tips have important implications for both measurement and policy.

Accurate income measurement is critical for measuring the income distribution and estimating poverty rates. In Basker, Foster, and Stinson (2024), we use the reported incomes of tipped full-service single-unit workers to estimate their 2018 poverty rate, replicating the official Census Bureau methodology used for poverty-rate estimation (Semega, Kollar, Creamer, and Mohanty, 2020). We estimate a poverty rate of 11.7% for this group, more than double the poverty rate for all workers, which was 5.1% that year. If these workers' incomes are mismeasured, their poverty rate may be somewhat—or much—lower than this calculation implies.

From a policy perspective, our results underscore that the tip credit encourages tip reporting, but only up to a point. A higher tip credit would likely increase tip reporting, as we see in the examples of New York, which increased its tip credit from 2014 to 2015, and Texas, where the tip credit increased from 58% in 2005 and 2006 to 71% starting in mid-2009. A higher tip credit could shift the entire distribution of tips, as we observed in the cases of New York and Texas, but it would not address the root cause of under-reporting. In addition, a higher tip credit could lead to pressure on more workers to over-report their tips.[18]

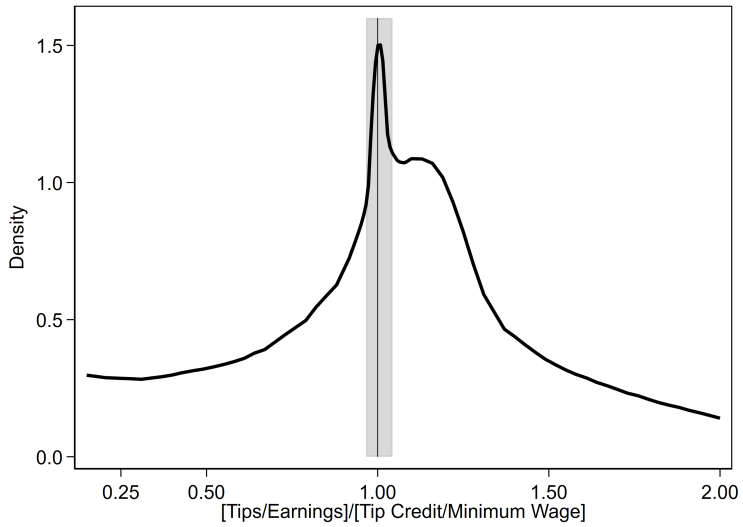
Other policy proposals may have a larger effect on tip reporting. The increase in tip

[18]See Bernhardt, Milkman, Theodore, Heckathorn, Auer, DeFilippis, González, Narro, Perelshteyn, Polson, and Spiller (2009) for a discussion of wage theft in the context of restaurant workers.

reporting since the introduction of third-party credit-card reporting in 2011 and the general shift towards electronic payments is consistent with these policies incentivizing higher tip reporting. In addition, starting in tax year 2025, tips up to $25,000 are exempt from federal income tax for many workers (Internal Revenue Service, 2026), reducing tipped workers' disincentive to report tips—at least up to that amount. This incentive change, however, is partial—payroll tax and state income tax are still collected on tip income—and may create a new bunching point for workers who earn more than $25,000 in tips.

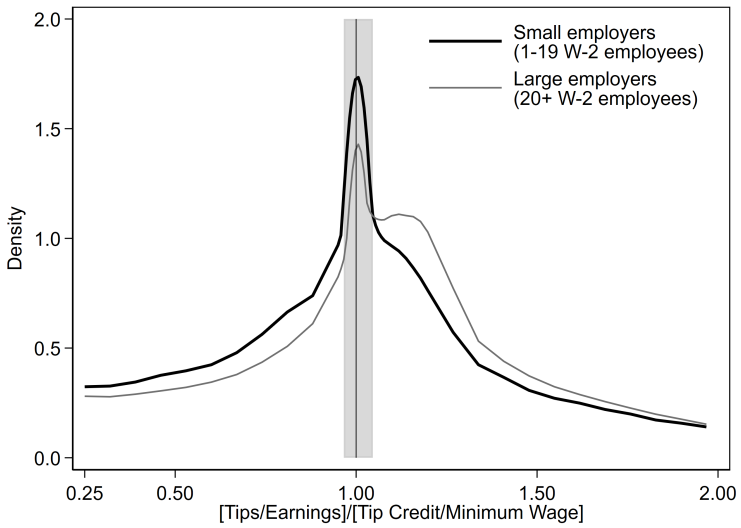
Finally, our finding that under-reporting of tips due to bunching at the tip-credit threshold accounts for a small fraction of overall tip under-reporting has important implications well beyond this study. The literature on bunching has focused on incentives that tax kinks and tax notches provide for individuals choosing hours of work, demonstrating how these rules change worker behavior and reporting. But in settings where there is a strong incentive for workers to under-report earnings, measuring behavior is difficult not only at thresholds; under-reporting throughout the distribution masks true underlying earnings. Even if bunching is estimated precisely, more data that provide an alternative estimate of the true income distribution—in our case, restaurant revenue—are needed to determine the full extent of the missing earnings. As our results demonstrate, the tip-credit threshold, while quantitatively important, represents only the tip of the iceberg of unreported tip income.

Figure 1. Distribution of Normalized Tips-to-Earnings Ratios among Tipped Jobs at Full-Service Single-Unit Restaurants Subject to a Tip Credit, 2005–2018



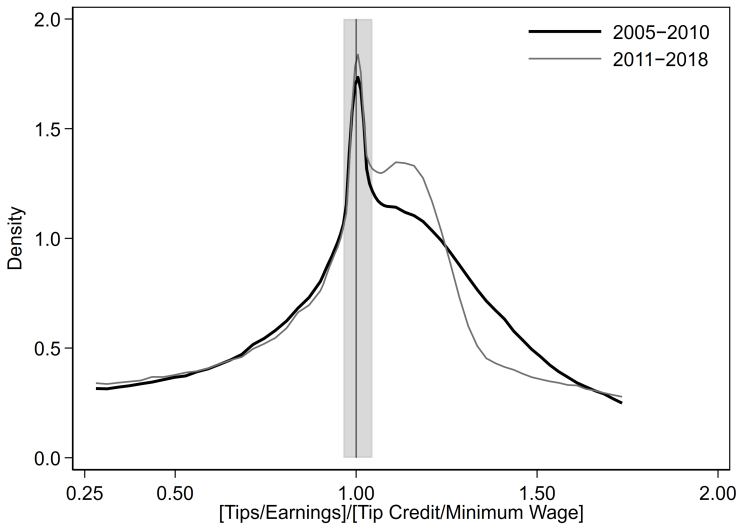
Notes: The figure shows the distribution of normalized tips-to-earnings ratios for a sample of 27.5 million job-years in states with tip credits (representing 9.5 million unique workers). This ratio is 1 when a worker in a job subject to a two-tiered minimum wage and tip credit reports exactly the tips to meet the tip credit. The shaded area is the bunching window: $[0.96, 1.05]$, with a vertical line at the reporting threshold of 1. Authors' calculations from Form W-2 and state minimum wages. Tails removed to comply with Census Bureau disclosure-avoidance rules.

Figure 2. Distribution of Normalized Tips-to-Earnings Ratios
by Employer Size, 2005–2018



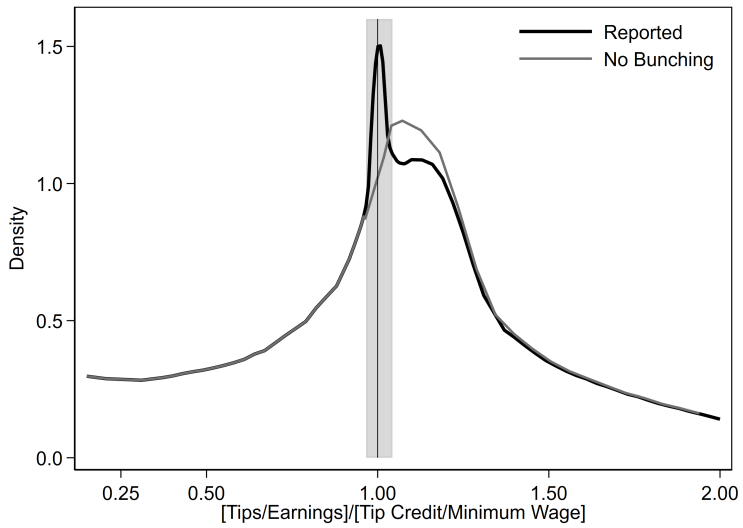
Notes: The figure shows the distribution of normalized tips-to-earnings ratios by employer size (small versus large) for a sample of 27.5 million job-years in states with tip credits (representing 9.5 million unique workers). This ratio is 1 when a worker in a job subject to a two-tiered minimum wage and tip credit reports exactly the tips to meet the tip credit. The shaded area is the bunching window: $[0.96, 1.05]$, with a vertical line at the reporting threshold of 1. Authors' calculations from Form W-2 and state minimum wages. Tails removed to comply with Census Bureau disclosure-avoidance rules.

Figure 3. Distribution of Normalized Tips-to-Earnings Ratios
for Two Periods, 2005–2010 and 2011–2018



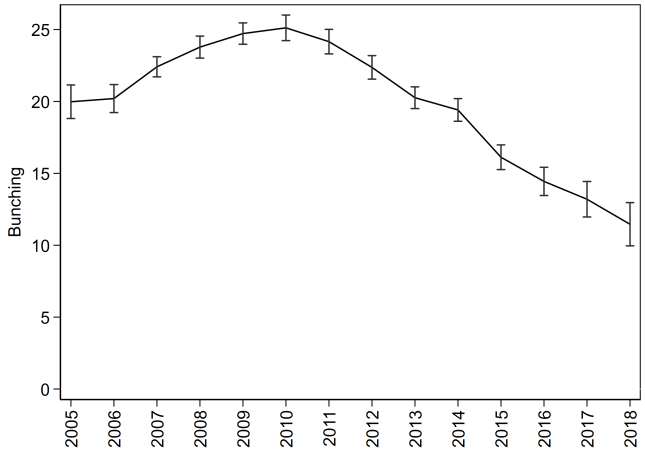
Notes: The figure shows the distribution of normalized tips-to-earnings ratios for two periods (2005–2010 and 2011–2018) for a sample of 27.5 million job-years in states with tip credits (representing 9.5 million unique workers). This ratio is 1 when a worker in a job subject to a two-tiered minimum wage and tip credit reports exactly the tips to meet the tip credit. The shaded area is the bunching window: $[0.96, 1.05]$, with a vertical line at the reporting threshold of 1. Authors' calculations from Form W-2 and state minimum wages. Tails removed to comply with Census Bureau disclosure-avoidance rules.

Figure 4. Distribution of Normalized Tips-to-Earnings Ratios and No-Bunching Counterfactual, 2005–2018



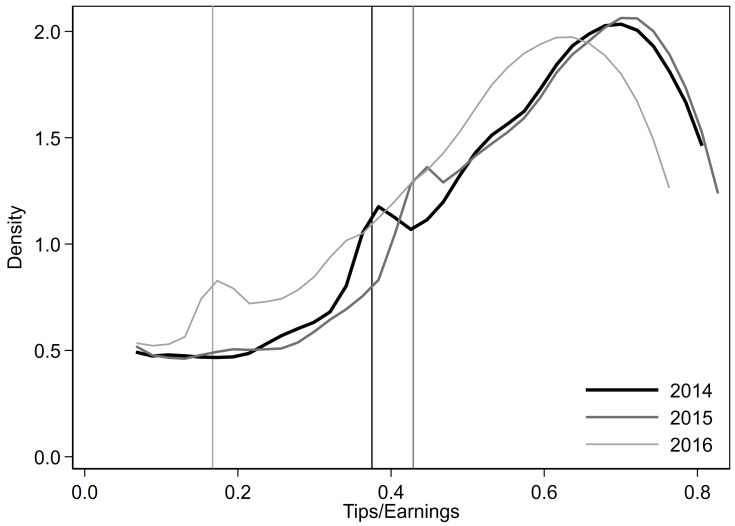
Notes: The figure shows the distribution of normalized tips-to-earnings ratios and the no-bunching counterfactual for a sample of 27.5 million job-years in states with tip credits (representing 9.5 million unique workers). This ratio is 1 when a worker in a job subject to a two-tiered minimum wage and tip credit reports exactly the tips to meet the tip credit. The shaded area is the bunching window: $[0.96, 1.05]$, with a vertical line at the reporting threshold of 1. Authors' calculations from Form W-2 and state minimum wages. Tails removed to comply with Census Bureau disclosure-avoidance rules.

Figure 5. Excess Bunching of Normalized Tips-to-Earnings Ratios, by Year



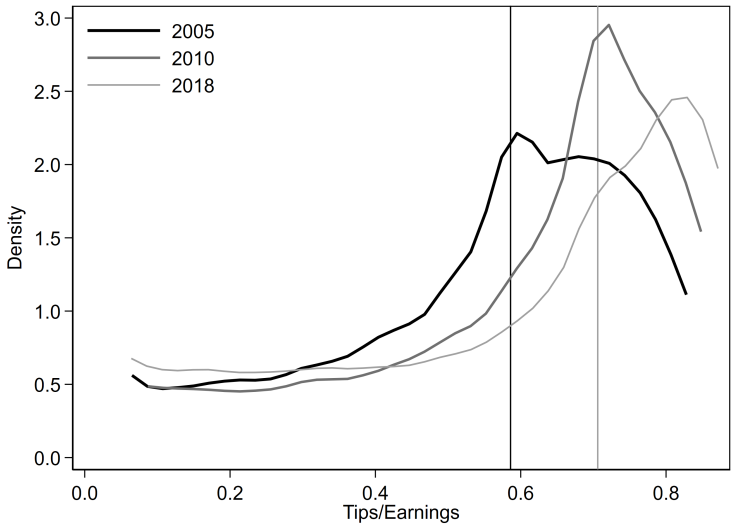
Notes: The figure shows excess bunching of normalized tips-to-earnings ratios for a sample of 27.5 million job-years in states with tip credits (representing 9.5 million unique workers). Bunching calculated as share of all jobs with normalized tips-to-earnings ratios in [0.96,1.05]. Vertical bars represent 95% confidence intervals. Authors' calculations from Form W-2 and state minimum wages.

Figure 6. Distribution of Tips-to-Earnings Ratios for New York (2014, 2015, and 2016)



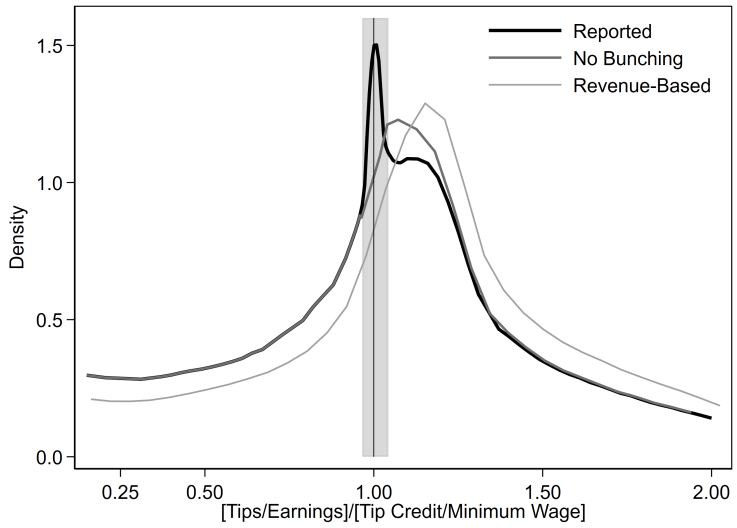
Notes: The figure shows the distribution of tips-to-earnings ratios for a sample of 668,000 job-years in New York (representing 331,000 unique workers). Vertical lines at 0.167 (2016 tip credit), 0.375 (2014 tip credit), and 0.429 (2015 tip credit). Authors' calculations from Form W-2 and state minimum wages. Tails removed to comply with Census Bureau disclosure-avoidance rules.

Figure 7. Distribution of Tips-to-Earnings Ratios for Texas (2005, 2010, and 2018)



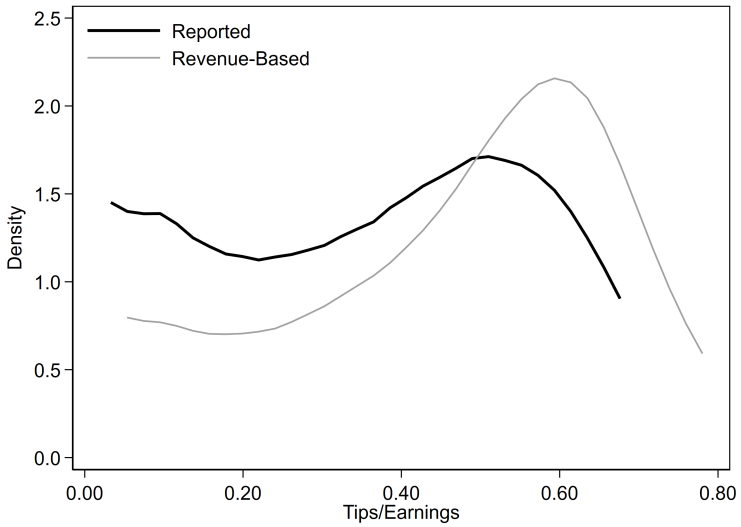
Notes: The figure shows the distribution of tips-to-earnings ratios for a sample of 484,000 job-years in Texas (representing 392,000 unique workers). Vertical lines at 0.586 (2005 tip credit) and 0.706 (2010 and 2018 tip credit). Authors' calculations from Form W-2 and state minimum wages. Tails removed to comply with Census Bureau disclosure-avoidance rules.

Figure 8. Distribution of Normalized Tips-to-Earnings Ratios, No-Bunching Counterfactual, and Revenue-Based Counterfactual, 2005–2018



Notes: The figure shows the distribution of normalized tips-to-earnings ratios, the no-bunching counterfactual, and the revenue-based counterfactual. This ratio is 1 when a worker in a job subject to a two-tiered minimum wage and tip credit reports exactly the tips to meet the tip credit. The shaded area is the bunching window: $[0.96, 1.05]$, with a vertical line at the reporting threshold of 1. Reported and no-bunching counterfactual use sample of 27.5 million job-years in states with tip credits (representing 9.5 million unique workers). Revenue-based counterfactual uses sample of 29.2 million job-years in states with tip credits (representing 10.4 million unique workers, including some workers who reported zero tips but worked at restaurants where tips were likely to be earned). Authors' calculations from Form W-2 (tips), BR/LBD (revenue), and state minimum wages. Tails removed to comply with Census Bureau disclosure-avoidance rules.

Figure 9. Distribution of Tips-to-Earnings Ratios and Revenue-Based Counterfactual for Non-Tip-Credit States (2005–2018)



Notes: The figure shows the distribution of tips-to-earnings ratios and revenue-based counterfactual for non-top-credit states comprised of a sample of 6.5 million job-years in Alaska, California, Hawaii, Minnesota, Montana, Nevada, Oregon, and Washington (representing 2.3 million unique workers). (Hawaii is a special case; see Section 2.) Authors' calculations from Form W-2 (tips) and BR/LBD (revenue). Tails removed to comply with Census Bureau disclosure-avoidance rules.

Table 1. Bunching by Employer Size, Tip-Credit States, 2005–2018

Sample	Mean	SD
All employers	19.82	0.270
Employers with 1–9 employees	30.28	0.869
Employers with 10–19 employees	30.19	0.577
Employers with 20–39 employees	26.39	0.353
Employers with 40 or more employees	14.89	0.249

Notes: The table shows bunching of normalized earnings-to-tips ratios for a sample of 27.5 million job-years in states with tip credits (representing 9.5 million unique workers) by employer size. Bunching calculated as share of all jobs with reported normalized earnings-to-tips ratios (defined as [Tips/Earnings]/[Tip Credit/Minimum Wage]) in [0.96,1.05]. Number of employees based on Form W-2.

Table 2. Reported and Missing Tips by Employer Size, Tip Status, and Minimum-Wage Regime, Annual Averages, 2005–2018

Sample	Reported Tips	Missing Tips	Share Missing
All employers, all states[a]	12.18	5.09	29%
Employers in tip-credit states[a,b]	9.85	3.67	27%
Employers with tipped workers[a]	9.85	2.68	21%
Employers with 1–9 employees	0.35	0.38	52%
Employers with 10–19 employees	0.85	0.55	39%
Employers with 20–39 employees	1.92	0.73	27%
Employers with 40 or more employees	6.73	1.03	13%
Employers without tipped workers	0	0.99	100%
Employers in non-tip-credit states[b]	2.33	1.42	38%

Notes: The table shows reported and missing tips in billions of real 2018 dollars for a sample of 2.63 million restaurant-years. Authors' calculations as follows. Restaurant reported tips aggregated from worker-level reports on Form W-2. Missing tips calculated based on the assumption that total tips aggregated over all workers must equal at least 12% of restaurant revenue. Revenue is from BR/LBD.

[a] May not equal sum of parts due to rounding.

[b] Non-tip-credit states include Alaska, California, Hawaii, Minnesota, Montana, Nevada, Oregon, and Washington. (Hawaii is a special case; see Section 2.) All other states are tip-credit states.

A Data Appendix

BR, LBD, and Form W-2

The BR is a list of all business establishments operating in the United States. The LBD tracks businesses across Employer Identification Number (EIN) changes over time; we make use of these longitudinal links whenever possible to mitigate the impact of re-organizations on our linking. For the purposes of this appendix, we refer to the two files as the BR/LBD.

We identify full-service restaurants primarily using NAICS codes from the BR.[19] These industry codes incorporate information shared with the Census Bureau by the IRS, Social Security Administration, and the Bureau of Labor Statistics (BLS) but are not based exclusively on administrative data. They are edited by Census Bureau staff to account for responses to various Census Bureau annual surveys as well as the Economic Census. All of these sources rely on businesses to describe their primary activity and are subject to reporting errors. Industry assignment is noisiest for single-unit firms, the majority of which are "non-mail" cases in the Economic Census, meaning they do not receive any Economic Census forms, so their data come entirely from administrative sources (Basker, 2021).

Given its hierarchical structure, a natural starting place to examine the accuracy of industry coding is at the two-digit level. Cuffe, Bhattacharjee, Etudo, Smith, Basdeo, Burbank, and Roberts (2022) find that 80% of single-unit establishments have the same industry code in Census Bureau and Social Security Administration records, and 86% have the same codes in Census Bureau and BLS records, but they do not provide a breakdown specifically for restaurants. Oehlert (2023) uses a sample of hand-validated IRS tax returns from 2012–2016 and reports that among businesses that reported sector 72, "Accommodation and Food Services," on their tax forms, the sector was correct in 92% to 96% of cases, depending on

[19] Although the NAICS code for full-service restaurants changed during the period of our sample, from 722110 before 2012 to 722511 in 2012 and thereafter, the definition of the industry did not change substantively. Non-full-service restaurants include fast-food (limited-service) restaurants, bars, coffee shops, food trucks, catering services, and more.

the form used (Oehlert, 2023, Tables 13 and 17). In our paper, depending on the year, we rely on either the fourth or sixth digit of the industry code to distinguish between full-service restaurants and other types of restaurant establishments (limited-service restaurants, bars, coffee shops, etc.). Elvery, Foster, Krizan, and Talan (2006) compare Census Bureau and BLS industry codes and find that, for single-unit firms, the rate of agreement falls from 85% at the sector level to 67% at the six-digit level, but they do not provide a breakdown for specific sectors.

BR/LBD revenue is primarily sourced from annual IRS business-income tax filings on Form 1120, Form 1065, Form 990, and Form 1040 Schedule C, depending on the business's legal form of organization. However, we cannot link all Form 941 tax filers to a revenue tax filing, likely because some firms use different tax identifiers on IRS payroll versus revenue forms: revenue information is missing for about 20% of businesses in the LBD (Haltiwanger, Jarmin, Kulick, and Miranda, 2017, p. 17). Unfortunately, revenue is not missing at random; it is much more likely to be missing for the very smallest of our full-service single-unit restaurants. In our sample, revenue is missing for about 30% of restaurant-year observations. In those cases, we impute revenue. We regress log revenue on indicators for establishment birth and death years, log count of workers, log payroll, and year and state fixed effects. We exponentiate the predicted value from this regression to impute the missing values.

The BR/LBD also includes information on aggregate tips, primarily sourced from IRS Form 941. Businesses file this tax form quarterly to remit payroll and income tax withheld from workers' earnings during the previous quarter and to pay the employer's share of the payroll tax. As explained below, we use Form 941 tips to validate Form W-2 in our analysis.

For more information about the BR/LBD, see Chow, Fort, Goetz, Goldschlag, Lawrence, Perlman, Stinson, and White (2021). For more information about revenue in the LBD, see Haltiwanger, Jarmin, Kulick, Miranda, Penciakova, and Tello-Trillo (2019) and Yeh and Macaluso (2023).

Form W-2 data are available by job-year, where a job is defined as a match between

a business and a person. These individual tax forms are filed by employers for all jobs for which income or payroll taxes are withheld, which includes virtually all restaurant jobs (see Internal Revenue Service, 2006b, 2019a).[20] We drop a small number of workers whose identifiers are associated with more than 24 unique employers in a given year. The IRS requests that Form W-2 and Form 941 be filed using the same EIN so that IRS can ensure that the amounts match across the two forms (Internal Revenue Service, 2006b, 2019a). Because of this, we can match most individual Form W-2 records to BR/LBD restaurant records. However, there are cases in which we do not match the two sources. These could be the result of late or missing filings or changes in EIN due to change in ownership. In our analysis of restaurant-level reporting, we treat any reporting of tips—whether on Form W-2 for a worker at the business, or on Form 941 for the business as a whole—as evidence that the business is reporting tips to the IRS.

IRS rules require workers to report tip income to their employers, which are then responsible for entering the information on the workers' Form W-2, and on Form 941 quarterly filings at the business level. Tips are reported separately on Form W-2 because they are treated differently from wage earnings for the purpose of FICA taxes. Workers pay half of the payroll taxes on their tips. Employers pay the other half, but they are allowed to take a credit against their income tax for payroll taxes on tips that contribute to workers earning more than $5.15 an hour (U.S. Department of Treasury, Office of Tax Analysis, 2016). Because of this employer rule, wages subject to FICA tax are reported on Box 3 on Form W-2 (Social Security wages) and tips subject to FICA tax are reported in Box 7 (Social Security tips). Form W-2 data housed at the Census Bureau for thie period studied in this paepr do not include Box 7, so we infer tips by subtracting Form W-2 Box 3 income (Social Security wages) from total compensation, which includes Box 1 income (wages,

[20]The federal payroll tax, often referred to as FICA based on the Federal Insurance Contributions Act law that authorized it, funds the Social Security Old-Age, Survivor, and Disability Insurance program and Medicare.

tips, other compensation) and Box 12 income (deferred compensation, such as 401(k) and Health Savings Account contributions, and other compensation, such as moving expenses). This difference gives us the portion of total compensation not treated as Social Security wages and is a valid measure of tips for employees of restaurants as long as the worker's total earnings are less than the FICA taxable maximum, which varies by year. In 2018, the FICA maximum was $128,400, well above what most tipped employees earn from tips and wages combined.[21] We drop any job-year observation if Box 3 income is at that year's FICA maximum to avoid over-imputing tips to non-tipped jobs. In addition, our method of calculating tips combines regular tips (Box 7) and allocated tips, which are required when a restaurant's workers collectively report insufficient tips to meet its required revenue share (Box 8).[22] However, IRS tabulations for 2008–2018 show that allocated tips are well under 1% of total reported tips every year (Internal Revenue Service, 2017).

To verify our tip calculations from Form W-2, we check them against the business-level reported tips in Form 941. First, we correlate a binary variable that equals 1 if the business has any reported Form 941 tips with a binary variable that equals 1 if any jobs at the business reported tips on Form W-2; that correlation is 94%. Second, restricting our attention to businesses with reported tips on both Form 941 and Form W-2, we correlate the log Form 941 tips with the log of aggregated Form W-2 tips; that correlation is 96%.

Unemployment Insurance

One more source of information at the job level is the Longitudinal Employer-Household Dynamics (LEHD) data infrastructure, which captures state unemployment-insurance records. LEHD data include total earnings by job-quarter. We use this information to refine minimum-wage assignments. Minimum wages sometimes change during a calendar year, as when the

[21] For a history of FICA taxable maximum amounts see Social Security Administration (undated).

[22] As noted in Section 2, that threshold is typically 8%. However, restaurants may petition the IRS for a lower rate, which can be as low as 2% (Internal Revenue Service, 2006a). We do not know how often such petitions are made or granted.

federal minimum wage increased in July of 2007, 2008, and 2009. Because many jobs in the restaurant industry last less than a full calendar year, we use LEHD information to more precisely identify the effective minimum wage at the job level.

Most Form W-2 jobs can be matched by EIN and person identifier to state UI records. In some cases, however, employers may use different tax identifiers for reporting employment to their state UI systems and for reporting to the IRS, in which case we are unable to match earnings records across sources.[23] Because LEHD includes Massachusetts only starting in 2010 (Hyatt, McEntarfer, McKinney, Tibbets, and Walton, 2014), we exclude restaurants in Massachusetts from the analysis in this paper. For jobs that do not match to LEHD, we use the mean value of the minimum wage faced by the business in a calendar year. For more information about the LEHD data infrastructure, see Abowd, Stephens, Vilhuber, Andersson, McKinney, Roemer, and Woodcock (2009) and Vilhuber (2018).

Minimum Wages

Minimum wages apply at the business level, meaning that it is the location (and sometimes the size and industry) of the business, not the state of residence of the worker, that determines the applicable minimum wage.

We use monthly minimum-wage data at the state and local level from 2005 to 2018. We deflate each monthly value using the all-items CPI and then average by calendar year. Our starting point for the regular minimum wages is a monthly file from Basker and Khan (2016) with state-level minimum wages for the period 1993–2014. We extend the dataset by four years using the Department of Labor's (DOL) website (U.S. Department of Labor, Wage and Hour Division, undated a). Our starting point for the tipped, subminimum, wages is an annual file created by Jones (2016). We supplement this file using another DOL website to

[23]Goldschlag, Jarmin, Lane, and Zolas (2021, Appendix IA.2) report that, across the universe of Form W-2 records in 2005–2014, an LEHD record is missing in approximately 20% of cases; conversely, Form W-2 is missing for approximately 10% of LEHD observations.

create monthly records (U.S. Department of Labor, Wage and Hour Division, undated b). Even and Macpherson (2014) construct a similar dataset ending in 2013.

For both the minimum and subminimum wages, we supplement state-level files with information on city- and county-specific minimum-wage ordinances. In some cases these ordinances were repealed or litigated and annulled by the courts. We have done extensive searches in city codes and news reports to compile the start and, when applicable, end dates of these higher minimum wages. In all, we have found approximately 45 localities that imposed a higher minimum wage than their states at some point during the sample period.

Some states and localities have different minimum wages for different types of jobs or businesses. For example, New York has different minimum wages for different industries and occupations; we use the minimum wage for food-service jobs in the restaurant industry. Connecticut has two different tipped-subminimum wages: one for bartenders, and one for all other waitstaff. We use the waitstaff subminimum wage for our tip-credit calculations. West Virginia instituted a lower minimum wage for employers with five or fewer employees, starting in July 2006, and Ohio instituted two different minimum wages, one for businesses with under $250,000 in revenues and one for those with more than $250,000 in annual revenue, in 2007; the revenue threshold has changed annually since then. In these and similar cases, we use information on employment and revenue from the BR/LBD to assign the relevant minimum wages to each restaurant.

Although Washington bans tip credits, Seattle instituted a different minimum wage for tipped jobs starting in 2015 at restaurants belonging to chains with up to 500 employees worldwide (Seattle Office of Labor Standards, 2019). These jobs must earn tips but there is no minimum level of tips for the jobs to be eligible for the lower minimum wage. Hawaii has a tip credit, but it was very small: from 2005 through 2014, the tip credit was $0.25. In 2015, the tip credit increased to $0.50, but tipped employees must earn at least $7/hour in tips to be eligible for the $0.50–0.75 tip credit (Hawaii Department of Labor and Industrial Relations, 2014). The Hawaii tip credit increased to $0.75 in 2016. Because the Hawaii

tip credit is not equal to the amount workers must earn in tips, we include Hawaii in the non-tip-credit states. The District of Columbia passed a ballot initiative in 2022 that will eventually eliminate the tip credit (D.C. Law 24-281), and the Chicago City Council passed a similar measure in 2023 (Romeo, 2023).

Finally, whenever the federal minimum wage exceeds a state or locality's statutory minimum wage, or the state has no statutory minimum wage, the federal value applies.

B Minimum-Wage Scenarios

In this appendix, we provide a series of thought experiments to illustrate how increases in the tip credit due to a (standard) minimum-wage increase shift the distribution of the reported-tips-to-earnings ratio to the left. Throughout this appendix, we assume for simplicity that the starting minimum wage is \$10, with a subminimum wage of \$5 and a tip credit of \$5; we then increase the minimum wage to \$15, leaving the subminimum wage fixed at \$5, which increases the tip credit to \$10.

First, consider employee A, who earns the subminimum wage of \$5 and reports \$5 in tips. (She may, in fact, earn more in tips, but she reports \$5.) This worker is therefore right at the bunching point. When the minimum wage increases to \$15, she starts reporting \$10 in tips to meet the new tip credit. This leaves her at the bunching point.

Next, consider employee B, who, like employee A, earns the subminimum wage of \$5 and reports \$5 in tips. When the minimum wage increases to \$15, she increases her reported tips from \$5 to \$7—not enough to meet the new tip credit. Employee B's employer adds \$3 to her base pay, per FSLA requirements. She therefore moves from being at the bunching point to being below it.

Employee C earns the subminimum wage of \$5, and reports \$12 in tips, although the tip credit is just \$5. He is above the bunching threshold. When the minimum wage increases to

$15, employee C continues getting paid $5 by his employer, and continues reporting $12 in tips. He is still above the bunching point, but moves closer to it.

Employee D is a busser who earns only $2 in tips. His employer therefore must pay him $8 in base wages so that his tip-inclusive hourly earnings meet the minimum-wage requirement. This puts employee D below the bunching threshold. When the minimum wage increases to $15, employee D continues reporting $2 in tips; his base wage increases to $13. He remains below the bunching threshold, moving away from it.

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